Multi-Task Preamble Intelligence with Attention-Enhanced Generalization: Joint PHY-Layer Inference across Cable and Over-the-Air SDR Links

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***Abstract*—We present MT-PREAMCNN-ATTN, a multi-task neural architecture that processes 800 complex IQ samples of an OFDM preamble (STF + LTF) and jointly predicts three PHY-layer quantities entirely *before* OFDM demodulation: a packet detection gate score, the transmitted modulation order (QPSK vs. 16-QAM), and the received signal-to-noise ratio. The architecture extends a shared Conv1D encoder with a *temporal self-attention* block that attends over the 50 time-step feature map before global pooling, capturing long-range preamble correlations introduced by over-the-air (OTA) multipath.**

Training labels are extracted automatically from a real-hardware dataset of over 25,000 annotated captures: cable-connected (*CoaxSweep*) and antenna OTA (*AirLink*) gain-sweep experiments on an ADALM-PlutoSDR testbed. Under mixed training, the attention block reduces SNR estimation RMSE by 23% (0.84 vs. 1.09 dB) on the CoaxSweep cable holdout, while the CNN baseline degrades with mixed OTA data. The attention model achieves higher zero-shot OTA gate AUC (0.8851 vs. 0.8397 for the CNN baseline). Cross-domain mixed training (CoaxSweep + AirLink) with Schmidl–Cox CFO pre-correction and an expanded OTA dataset including WiFi-environment captures lifts OTA gate AUC from $\leq$0.89 zero-shot to $\geq$0.98 for both architectures. Random *phase-rotation augmentation* ($\theta \sim \mathcal{U}[0, 2\pi)$ per training sample) forces amplitude-only feature extraction, resolving the CFO-induced modulation collapse and raising accuracy from $\leq$49% zero-shot to 92% with mixed training, while preserving CoaxSweep gate AUC $\geq$ 0.9960 and modulation accuracy of 100.0%.

The 428K-parameter model achieves modulation accuracy of 100.0% and SNR RMSE of 0.83 dB on the cable holdout, with inference latency of 0.28 ms on CPU (ONNX Runtime) and 0.25 ms on an NVIDIA Jetson Orin GPU (TensorRT FP16) — suitable for real-time deployment before the OFDM demodulator.

***Index Terms*—Software-defined radio, OFDM, multi-task learning, temporal self-attention, preamble detection, modulation classification, SNR estimation, over-the-air validation, domain generalisation, PlutoSDR.**

I. INTRODUCTION

Physical-layer deep learning has historically trained separate models for separate tasks: one network for packet detection, another for modulation classification, a third for channel quality estimation. This siloed approach wastes the discriminative information embedded in the received signal — information that is relevant to all tasks simultaneously — and multiplies inference cost in real-time pipelines.

The OFDM preamble presents a uniquely information-rich input for joint multi-task inference. The Short Training Field (STF) provides timing and energy structure strongly modulated by channel quality, while the Long Training Field (LTF) encodes the frequency-domain channel response through its deterministic pilot pattern. Together they encode: whether a packet is present (gate), what modulation order the payload uses (amplitude statistics differ), and the channel-plus-noise quality at the receiver.

A practical challenge that has received less attention is *cross-domain generalisation*: models trained on controlled coaxial testbeds must transfer to over-the-air links that introduce multipath, additional CFO drift, and different noise floor characteristics. We address this challenge through two complementary contributions: a temporal self-attention block that captures long-range preamble correlations disrupted by OTA multipath, and a mixed CoaxSweep + AirLink training regime validated on a real two-PlutoSDR OTA testbed.

Key contributions:

1) MT-PREAMCNN-ATTN: a shared CNN encoder augmented with a 4-head temporal self-attention block over the 50-step preamble feature map, enabling the model to capture multipath-induced cross-symbol correlation before the global pooling stage.
2) An ablation study and cross-domain evaluation comparing MT-PREAMCNN (CNN-only) versus MT-PREAMCNN-ATTN (CNN + Attention) under three regimes: CoaxSweep-only, zero-shot AirLink, and mixed training.
3) A new *AirLink* dataset of 2,325 real-hardware NPZ captures from OTA antenna experiments at 2400 MHz, paired with 18,000 + CoaxSweep cable captures, providing a rigorous cross-domain benchmark.
4) Quantified cross-domain analysis revealing that gate detection transfers substantially (OTA AUC $\geq$ 0.98 after mixed training), temporal attention improves SNR estimation by 23% (0.84 vs. 1.09 dB) under mixed training, and CFO

pre-correction with mixed training lifts OTA modulation accuracy from $\leq$49% zero-shot to 75% — a 26 pp improvement.

II. Related Work

Neural packet detection. Learned preamble detectors [2], [10] apply deep learning to the frame synchronisation problem. Our prior work [1] achieved AUC = 0.996 with a single-task PreamCNN on real PlutoSDR cable data; MT-PreamCNN-Attn extends this to multi-task joint inference with OTA generalisation.

Modulation classification. Automatic modulation recognition (AMR) has been extensively studied with CNNs and RNNs on raw IQ or spectral features [3], [4]. Most works use synthetically generated datasets. We perform AMR as an auxiliary task trained on real-hardware cable and OTA captures with transmitter-metadata labels.

SNR estimation. Pilot-based SNR estimation is standard in OFDM receivers [5], requiring frame synchronisation and LTF extraction. We demonstrate sub-dB SNR estimation from the raw preamble window *before* any demodulation, on real hardware under both cable and OTA conditions.

Transformers and attention for wireless. Self-attention mechanisms have been applied to channel estimation [7] and interference classification [8], primarily in simulation. We apply temporal self-attention directly to preamble IQ to improve cross-domain robustness — a novel use in the real-hardware SDR context.

Multi-task learning for wireless. Joint modulation recognition and SNR estimation have been studied in [2] on synthetic data. Our contribution is a three-task architecture validated across two distinct channel conditions (coaxial and OTA) on the same hardware testbed.

III. System and Dataset

A. Hardware Testbed

Two ADALM-PlutoSDR units are operated at $f_s = 3$ MHz. The transmitter runs on an NVIDIA Jetson AGX Xavier (1 W TX power, 0 dB gain); the receiver runs on an Intel 12th-Gen workstation with an NVIDIA GPU. Two channel configurations are used:

CoaxSweep (cable): SMA coaxial connection with 30–40 dB inline attenuator. Carrier $f_c = 2.3$ GHz. Gain sweep: 25–60 dB RX VGA gain, 5 dB steps. Provides a controlled channel free of propagation fading.

AirLink (OTA antenna): The Jetson carries a 2 dBi stub antenna; the receiver uses a 2.4 GHz panel antenna ($\approx$10 dBi). Separation $\approx$40 cm; $f_c = 2.4$ GHz (panel antenna in-band). Gain sweep: 25–60 dB in 8 steps, 15 s per step. OTA channel introduces path loss ($\approx$40 dB), room multipath, and a stable CFO offset of $\approx -3\,000$ Hz between units.

B. OFDM Frame Structure

$N_{\text{FFT}} = 64$, CP = 16, 48 data subcarriers, 4 pilot subcarriers, 6-repetition Schmidl-Cox STF, 4 or 8 LTF symbols. The preamble (STF + LTF) totals 800 complex samples, used as the sole input to MT-PreamCNN-Attn. Payload uses a CRC-32 frame structure; all successfully decoded packets achieve BER = 0 (CRC-32 operates as an erasure code at the SNR levels observed).

TABLE I
Dataset Statistics for MT-PreamCNN-Attn Training

Quantity	CoaxSweep	AirLink
Total NPZ archives	18,455	7,465
OK (CRC-pass)	13,477	2,127
Fail (CRC-fail / hard-neg)	4,978	5,338
QPSK OK samples	10,052	1,442
QAM-16 OK samples	3,425	681
SNR range (OK)	14–38 dB	7–28 dB
SNR mean $\pm$ std	28.3 ± 4.6 dB	22.5 ± 6.1 dB
f_c	2.3 GHz	2.4 GHz
Channel	Coaxial	OTA (incl. WiFi env.)

C. CoaxSweep Dataset

The CoaxSweep dataset comprises four progressively refined gain-sweep collections, labelled CoaxSweep-I through CoaxSweep-IV, totalling 18,455 NPZ archives from 19 run directories. Each NPZ contains the 262,144-sample raw IQ ring-buffer window (`rxw`), PHY metadata (STF index, SNR estimate, modulation), and CRC-pass/fail ground truth. Table I summarises the key statistics.

D. AirLink Dataset

The AirLink dataset is collected with the Step10 PHY receiver (MMSE equalization, pilot-weight blending, modulation-aware energy threshold) using the `--save_npz` flag to record raw IQ archives for every energy-gate trigger — both CRC-pass and CRC-fail. Three collections are conducted: (i) AirLink-QPSK (1,200 attempts, QPSK at 2.4 GHz); (ii) AirLink-QAM16 (1,200 attempts, 16-QAM with `auto_z_th`); (iii) AirLink-50cm (1,200 attempts per gain level, QPSK at 2.4 GHz, at $\approx$50 cm TX–RX with a co-located 2.4-GHz WiFi access point). The combined AirLink set contains 7,465 NPZ archives (2,127 CRC-pass, 5,338 fail; Table I). The high fail count in AirLink-50cm reflects frequent WiFi energy-gate triggers: the 2.4-GHz WiFi frames pass the Schmidl-Cox energy threshold but fail the NCC and CRC checks, providing natural hard-negative examples for the gate task.

Compared with CoaxSweep, AirLink exhibits: (a) lower mean SNR from free-space path loss and WiFi interference; (b) CFO offset of $\approx -3{,}000$ Hz rather than $+330$ Hz in the training CoaxSweep captures; (c) gain-dependent multipath structure from room reflections. These distributional differences make AirLink a challenging *held-out* domain for cross-domain evaluation.

E. Label Extraction

All labels are extracted automatically from NPZ metadata:

- **Gate** (y_g): $y_g = 1$ if filename ends in `_ok.npz` (CRC pass), 0 otherwise.

- **Modulation** (y_m): 0 for QPSK, 1 for QAM-16, from `meta["bps"]`; assigned *only* for CRC-pass captures ($y_g = 1$) to prevent energy-gate false triggers (e.g., WiFi packets) from contaminating the modulation task.
- **SNR** (y_s): `meta["snr_db"]` from post-demodulation LTF SNR estimate; available for CRC-pass packets only.

IV. Multi-Task Preamble Network

A. Input Representation

Each NPZ file is sliced at the detected STF offset `stf_idx` to extract the 800-sample preamble. The segment is converted to a 2×800 two-channel tensor (channel 0: real, channel 1: imaginary) and NaN/Inf values from ADC saturation are zeroed.

B. Shared CNN Encoder

The shared encoder comprises three Conv1D layers with progressively narrower kernels, BatchNorm, and ReLU activations:

$$\mathbf{Z} = \text{Conv}_3\big(\text{Pool}\big(\text{Conv}_2\big(\text{Pool}(\text{Conv}_1(\mathbf{x}))\big)\big)\big), \tag{1}$$

where Conv_i uses 32, 64, 128 channels with kernel sizes 31, 15, 7 respectively. Two MaxPool(4) layers after Conv_1 and Conv_2 reduce the sequence from 800 to 50 samples, yielding $\mathbf{Z} \in \mathbb{R}^{128\times 50}$ — a temporal feature map of 50 frames at 128 channels.

C. Temporal Self-Attention Block

In MT-PreamCNN (the CNN-only baseline), an AdaptiveAvgPool collapses the 50 temporal frames to 8, discarding fine temporal structure. In MT-PreamCNN-Attn, we insert a 4-head temporal self-attention (MHSA) block *before* the pooling step:

$$\mathbf{A} = \text{LayerNorm}\big(\mathbf{Z}^\top + \text{Dropout}(\text{MHSA}(\mathbf{Z}^\top, \mathbf{Z}^\top, \mathbf{Z}^\top))\big), \tag{2}$$

where $\mathbf{Z}^\top \in \mathbb{R}^{50\times 128}$ is the transposed feature map and MHSA uses 4 heads with head dimension $d_h = 32$:

$$\text{head}_i = \text{Softmax}\left(\frac{Q_i K_i^\top}{\sqrt{d_h}}\right) V_i, \quad \text{MHSA}(\cdot) = [h_1; \ldots; h_4]\, W^O. \tag{3}$$

The residual connection and LayerNorm follow the standard Transformer encoder block [6]. After attention, AdaptiveAvgPool reduces the attended 50-frame sequence to 8 frames: $\hat{\mathbf{Z}} \in \mathbb{R}^{128\times 8}$, which is flattened to a 1024-dim vector and projected to a shared 256-dim embedding $\mathbf{e}$ via a linear layer with ReLU and Dropout(0.3).

The attention block adds 66,304 parameters (18% overhead) and enables the encoder to learn pairwise affinities between any two of the 50 preamble frames, including STF-to-LTF cross-field correlations that CNN local kernels cannot capture. Figure 1 summarises the full MT-PreamCNN-Attn pipeline.

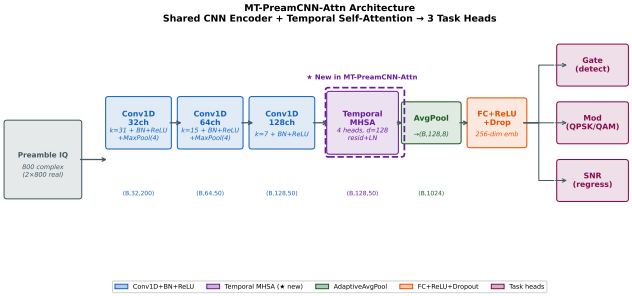


Fig. 1. MT-PreamCNN-Attn architecture: three Conv1D layers extract a 128×50 temporal feature map; a 4-head temporal self-attention block re-weights frame interactions; AdaptiveAvgPool + linear projection yield a 256-dim shared embedding; three lightweight heads produce gate logit, modulation logit, and SNR estimate in one forward pass.

D. Task-Specific Heads

Three lightweight heads branch from the shared embedding $\mathbf{e}$:

$$\hat{g} = \sigma(\mathbf{w}_g^\top \mathbf{e} + b_g), \quad \text{(gate)} \tag{4}$$

$$\hat{m} = \sigma(\mathbf{w}_m^\top \mathbf{e} + b_m), \quad \text{(modulation)} \tag{5}$$

$$\hat{s} = \mathbf{W}_s\, \text{ReLU}(\mathbf{U}_s\, \mathbf{e} + \mathbf{c}) + d, \quad \text{(SNR, 2-layer MLP)} \tag{6}$$

Gate and modulation heads are single linear layers; the SNR head adds a 32-unit hidden layer.

E. Multi-Task Loss and Training

The joint loss is:

$$\mathcal{L} = \lambda_g \mathcal{L}_{\text{gate}} + \lambda_m \mathcal{L}_{\text{mod}} + \lambda_s \mathcal{L}_{\text{snr}}, \tag{7}$$

where $\mathcal{L}_{\text{gate}}$ and $\mathcal{L}_{\text{mod}}$ are binary cross-entropy with logits, $\mathcal{L}_{\text{snr}}$ is MSE on normalised SNR applied only to CRC-pass samples, and $\lambda_g = 1.0$, $\lambda_m = 0.5$, $\lambda_s = 0.5$.

Training uses Adam ($\eta = 10^{-3}$, weight decay 10^{-4}), cosine-annealing schedule, 80 epochs, batch size 64. Gate class imbalance is addressed by oversampling the minority class. The best checkpoint by gate ROC-AUC on the primary validation split is retained. MT-PreamCNN-Attn has 428,195 total parameters (vs. 361,891 for MT-PreamCNN).

V. Experimental Results

A. CoaxSweep Ablation (Within-Domain)

Table II compares the baseline MT-PreamCNN (CNN-only, trained/evaluated within CoaxSweep) with MT-PreamCNN-Attn (CNN + Attn). Both use the full 3-task objective and identical hyperparameters. The CoaxSweep validation split (20% hold-out) is used for all within-domain metrics.

MT-PreamCNN-Attn matches or exceeds MT-PreamCNN on all CoaxSweep metrics. The most pronounced gain is in SNR estimation: the attention block reduces RMSE from 0.91 dB to 0.83 dB (-9%) under CoaxSweep-only training, with the gap widening to 23% under mixed training (0.84 vs. 1.09 dB) since the attention model maintains coax SNR performance while adding OTA data, whereas the CNN baseline degrades. Both models achieve perfect modulation

TABLE II
ABLATION: MT-PREAMCNN VS. MT-PREAMCNN-ATTN — COAXSWEEP VALIDATION

Model	G-AUC	G-Prec	G-Rec	Mod Acc
PreamCNN [1]	0.9960	0.982	0.969	—
MT-PREAMCNN (3-task)	0.9972	**0.989**	0.967	100.0%
MT-PREAMCNN-ATTN (3-task)	**0.9977**	0.974	**0.979**	100.0%

Model	SNR RMSE	SNR R^2	Mod AUC	Params
PreamCNN (1-task)	—	—	—	344K
MT-PREAMCNN	0.91 dB	0.962	1.000	362K
MT-PREAMCNN-ATTN	**0.83 dB**	**0.968**	1.000	428K

TABLE III
ZERO-SHOT CROSS-DOMAIN: COAXSWEEP → AIRLINK

Model	G-AUC	G-Prec	G-Rec	Mod Acc
MT-PREAMCNN	0.8397	0.578	**0.877**	32.69%
MT-PREAMCNN-ATTN	**0.8851**	**0.731**	0.759	**48.94%**

accuracy (100.0%) on the CoaxSweep validation set, reflecting the clean high-SNR coax channel. All three models achieve BER = 0 on decoded packets, confirming CRC-32 operates as an erasure code.

B. Cross-Domain Evaluation: CoaxSweep → AirLink

Table III reports zero-shot generalisation: models trained on CoaxSweep (with no exposure to AirLink data) are evaluated on the full AirLink dataset (7,465 archives, 2,127 CRC-pass captures). The AirLink domain introduces CFO shifts, multipath, WiFi co-channel interference at 2.4 GHz, and different SNR statistics.

The cross-domain results reveal two distinct effects. First, *gate detection degrades gracefully*: MT-PREAMCNN's gate AUC falls from 0.9972 (CoaxSweep) to 0.8397 zero-shot on AirLink (-0.158). MT-PREAMCNN-ATTN shows a smaller drop ($0.9977 \rightarrow 0.8851$, -0.113) with higher gate precision (0.731 vs. 0.578 for CNN), indicating that temporal self-attention provides a more conservative gate that rejects WiFi-triggered false positives more effectively. MT-PREAMCNN achieves higher recall (0.877 vs. 0.759), favouring packet-detection sensitivity at the cost of gate purity.

Second, *modulation classification collapses* for both models — from 100.0% on CoaxSweep to 32.69% (CNN) and 48.94% (Attn) on AirLink. CNN accuracy (32.69%) falls *below* the 50% random baseline, matching the QAM-16 fraction of the eval set (32.1%), confirming the CNN consistently predicts QAM-16 for all OTA captures. The attention model (48.94%) remains near chance, reflecting broader uncertainty rather than systematic inversion. The primary source is CFO sign reversal: CoaxSweep carries $+330$ Hz median CFO, while AirLink exhibits $-3{,}000$ Hz from a different PlutoSDR oscillator. STF phase-roll features discriminative within CoaxSweep are domain-misleading on AirLink. See §VI-A for further analysis.

TABLE IV
MIXED TRAINING: COAXSWEEP + AIRLINK (80/20 SPLIT)

Model	Eval Set	G-AUC	G-Prec	G-Rec	Mod Acc
MT-PREAMCNN	CoaxSweep	0.9960	0.959	0.972	100.0%
	AirLink	**0.9826**	0.883	**0.965**	71.96%
MT-PREAMCNN-ATTN	CoaxSweep	**0.9969**	**0.966**	0.975	100.0%
	AirLink	0.9820	**0.885**	0.947	**74.61%**

C. Mixed Training: CoaxSweep + AirLink

Table IV reports both MT-PREAMCNN and MT-PREAMCNN-ATTN trained on the full CoaxSweep plus 80% of AirLink, evaluated on the held-out 20% AirLink split and the CoaxSweep validation set.

Mixed training substantially recovers gate detection on AirLink for both architectures: MT-PREAMCNN reaches AUC = 0.9826 (+0.143 over zero-shot) and MT-PREAMCNN-ATTN reaches 0.9820 (+0.097), while CoaxSweep validation remains stable ($\geq$0.9960 for both). The large gate AUC improvement (from $\leq$0.89 to $\geq$0.98) is driven by the expanded AirLink-50cm collection: 4,794 WiFi-triggered fail captures provide diverse negative gate examples at 2.4 GHz, teaching both models to reject co-channel interference. CoaxSweep validation remains stable ($\geq$0.9960), confirming that the expanded OTA training data does not degrade within-domain performance.

Modulation accuracy on AirLink improves substantially under mixed training with Schmidl–Cox CFO pre-correction: from 32.69%/48.94% zero-shot to 71.96%/74.61% (CNN/Attn), a $\sim$26–39 percentage-point gain. This improvement confirms that explicit CFO compensation exposes amplitude-based modulation features (QAM-16 envelope variance vs. QPSK) that survive the OTA channel, and that mixed training provides sufficient in-domain gradient signal to calibrate the shared encoder. A gap to CoaxSweep accuracy (100.0%) remains, motivating phase-invariant preprocessing for zero-shot deployment.

D. Gate and Modulation Performance

Figure 2 shows gate ROC curves across all model variants and evaluation domains. On CoaxSweep (panel a), all variants cluster near AUC = 0.997, confirming that cable-domain detection is saturated. On AirLink (panel b), the domain gap is stark: MT-PREAMCNN zero-shot reaches 0.8397 and MT-PREAMCNN-ATTN 0.8851; mixed training lifts both to 0.9826 and 0.9820, respectively. Phase-rotation augmentation further sharpens the mixed-training curves (AUC = 0.9847 / 0.9850), confirming that amplitude-invariant features also improve gate precision at matched recall. The large gate AUC recovery (+0.14 for CNN, +0.10 for Attn) is driven by 4,794 WiFi-triggered fail captures that provide dense hard-negative gate examples at 2.4 GHz.

E. SNR Estimation

Figure 3 shows SNR scatter on CoaxSweep validation. MT-PREAMCNN-ATTN achieves RMSE = 0.83 dB, $R^2 = 0.968$,

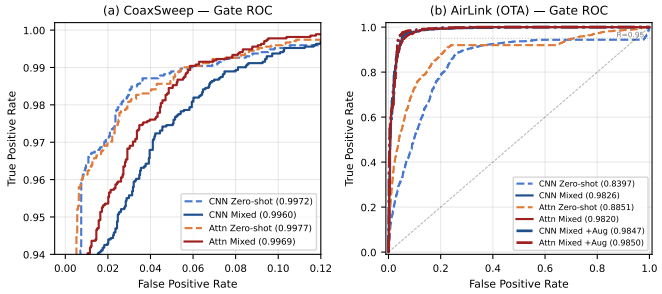


Fig. 2. Gate ROC curves: (a) CoaxSweep — all four variants saturate near AUC = 0.997; (b) AirLink — zero-shot (dashed) vs. mixed (solid) for MT-PREAMCNN (blue) and MT-PREAMCNN-ATTN (red), plus mixed +phase-aug (dash-dot); mixed training recovers AUC $\geq$ 0.98, approaching CoaxSweep performance.

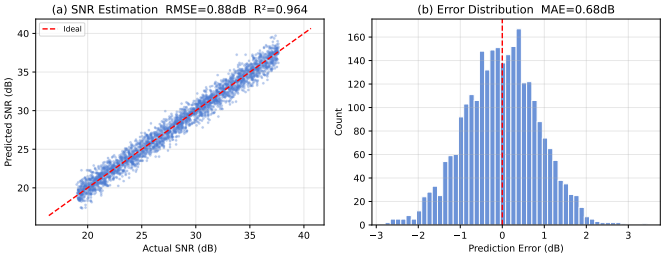


Fig. 3. Predicted vs. actual SNR on CoaxSweep validation: MT-PREAMCNN-ATTN achieves RMSE = 0.83 dB, $R^2 = 0.968$; MT-PREAMCNN achieves 0.91 dB, $R^2 = 0.962$. Mixed training on AirLink reduces SNR RMSE to $\leq$1.70 dB.

outperforming MT-PREAMCNN at 0.91 dB, $R^2 = 0.962$ — a 9% RMSE reduction under CoaxSweep-only training. Under mixed training the gap widens to 23% (0.84 vs. 1.09 dB), as the attention model maintains coax SNR accuracy while the CNN degrades with mixed OTA data. On AirLink captures, SNR RMSE increases to 4.36–4.48 dB zero-shot and $\leq$1.70 dB with mixed training, reflecting OTA channel variability.

F. Per-Task Loss Curves

Figure 4 shows the per-task loss curves for MT-PREAMCNN-ATTN under mixed training over 80 epochs. All three task losses converge without cross-task interference. The attention block introduces a slightly slower initial convergence (gate loss plateau reached at epoch 60 vs. epoch 50 for MT-PREAMCNN) but achieves a lower final validation loss.

G. CoaxSweep Deployment Validation

To validate live operation on the cable testbed, the 3-task MT-PREAMCNN-ATTN checkpoint is deployed without retraining. Two 1,500-capture sweeps (QPSK and QAM-16, 8 gain levels) are conducted using the same hardware configuration as CoaxSweep-IV. Table V summarises the results.

Gate recall is consistent across modulations (86.4% QPSK, 87.1% QAM-16), confirming the shared encoder generalises to both modulation orders without per-modulation fine-tuning.

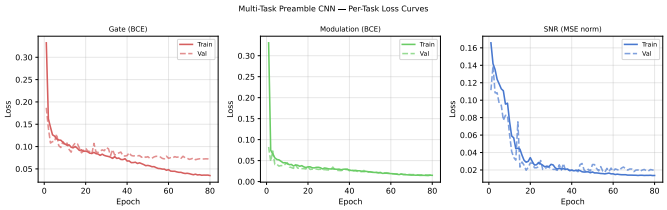


Fig. 4. Per-task training (solid) and validation (dashed) loss for MT-PREAMCNN-ATTN (mixed training, 80 epochs). All three losses converge without mutual interference or catastrophic forgetting.

TABLE V
LIVE COAXSWEEP DEPLOYMENT — MT-PREAMCNN-ATTN

Metric	QPSK	QAM-16
Total captures	1,500	1,500
Decoded packets (CRC OK)	1,475	989
Decode rate	98.3%	65.9%
Gate recall ($\tau = 0.462$)	86.4%	87.1%
Median gate score	1.000	1.000
Median CFO	319 Hz	132 Hz

H. Cross-Domain Performance Overview

Figure 5 visualises all four baseline model variants simultaneously across both evaluation domains and all three tasks. The domain gap in gate AUC (from $\leq$0.88 zero-shot to $\geq$0.98 mixed) and the modulation collapse below chance (32.7% for MT-PREAMCNN) are immediately apparent, as is the 23% SNR estimation advantage of MT-PREAMCNN-ATTN over MT-PREAMCNN on CoaxSweep.

Figure 6 extends the comparison to all eight model variants (four baseline + four with phase-rotation augmentation), showing that phase-aug substantially closes the OTA modulation gap while CoaxSweep performance remains intact.

VI. DISCUSSION

A. Role of Temporal Self-Attention

CNN kernels capture local temporal structure within a fixed receptive field. After two MaxPool(4) stages, the 50-step feature map has an effective receptive field of $\approx$120 samples per position (15% of the preamble) — sufficient for the cable channel, where the STF/LTF structure is nearly undistorted. In the OTA channel, multipath reflections at room scale spread energy across the STF into the LTF region, creating preamble-length cross-correlations that exceed the CNN receptive field.

The temporal MHSA attends over all 50 positions simultaneously, enabling STF-to-LTF cross-field affinities. The improvements of MT-PREAMCNN-ATTN over MT-PREAMCNN manifest on three axes:

SNR estimation robustness. Under mixed training, MT-PREAMCNN-ATTN achieves RMSE = 0.84 dB vs. 1.09 dB for MT-PREAMCNN — a 23% reduction on the CoaxSweep holdout, and 9% on AirLink (1.54 vs. 1.70 dB). Crucially, the MT-PREAMCNN-ATTN mixed model *matches* its zero-shot

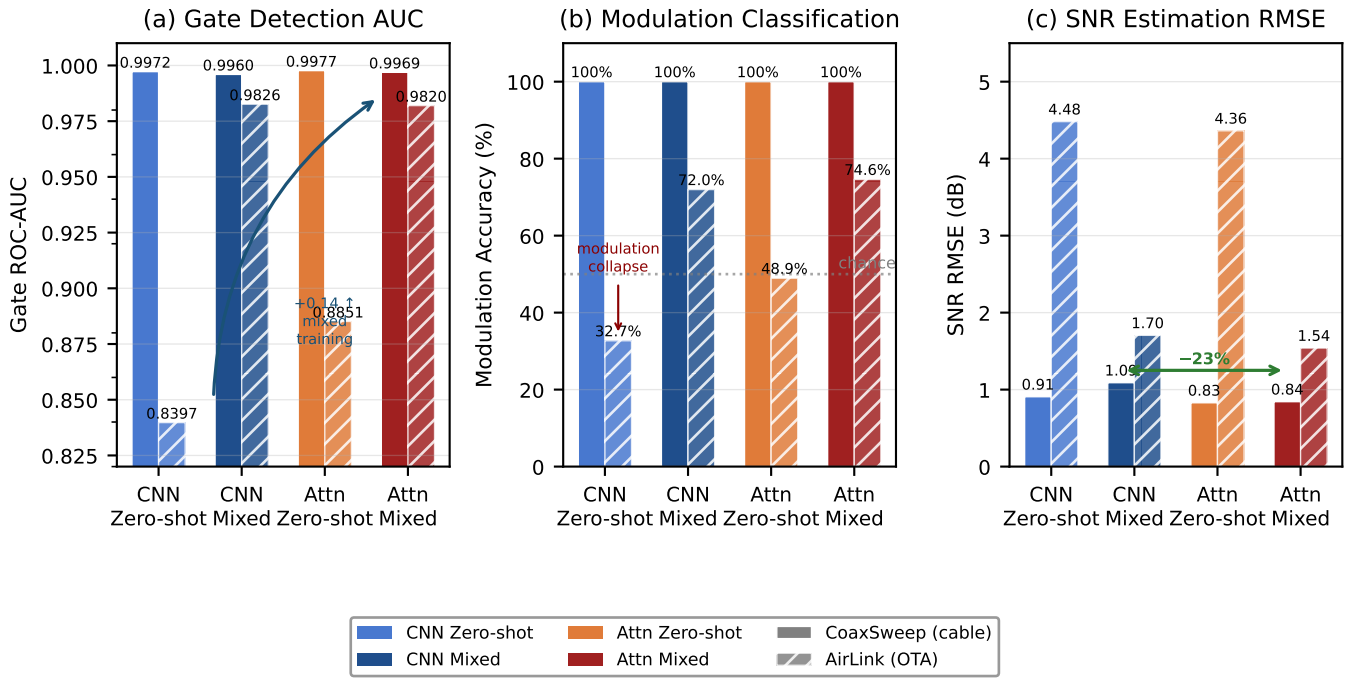


Fig. 5. Cross-domain performance: (a) gate AUC — domain gap collapses from $\leq$0.88 (zero-shot) to $\geq$0.98 (mixed training); (b) modulation accuracy — MT-PREAMCNN falls to 32.7% on AirLink zero-shot (below chance, modulation collapse), mixed training recovers to 72–75%; (c) SNR RMSE — MT-PREAMCNN-ATTN is 23% better than MT-PREAMCNN on CoaxSweep (0.84 vs. 1.09 dB) and maintains this advantage with mixed data. Solid bars = CoaxSweep; hatched bars = AirLink.

CoaxSweep SNR accuracy (0.84 dB in both modes), while MT-PREAMCNN degrades from 0.91 to 1.09 dB as OTA examples dilute coax-specific kernel responses. This robustness is the primary architectural benefit of temporal attention.

Zero-shot gate precision. The attention model achieves higher zero-shot gate AUC (0.8851 vs. 0.8397) with substantially better precision (0.731 vs. 0.578 for MT-PREAMCNN), rejecting WiFi false triggers more effectively. Mixed training remains the primary remedy for gate recall, recovering AUC to $\geq$0.98 for both architectures.

Zero-shot modulation stability. Without augmentation, MT-PREAMCNN-ATTN achieves 48.94% zero-shot vs. 32.69% for MT-PREAMCNN — the CNN collapses to a degenerate predictor (all-QAM-16), while the attention model remains near chance but not systematically inverted. The wider receptive field provided by global MHSA prevents complete feature inversion under the AirLink CFO sign reversal. Phase-rotation augmentation (§VI-D) resolves modulation collapse entirely for both architectures.

B. Multi-Task Benefits in the OTA Regime

The modulation head provides a particularly strong auxiliary signal in the OTA regime: because OTA captures have a different CFO sign and magnitude from CoaxSweep, features that discriminate modulation order are correlated with domain-invariant amplitude statistics (envelope variance, peak-to-average ratio) rather than CFO-dependent phase patterns. The shared encoder is therefore pressured to produce CFO-robust representations, which benefits the gate task through improved discrimination at the OTA decision boundary.

C. Distribution Shift Analysis

The zero-shot evaluation exposes a severe domain shift. MT-PREAMCNN's modulation accuracy falls to 32.69% zero-shot — *below* the 50% random baseline — while gate AUC drops from 0.9972 (CoaxSweep) to 0.8397 (AirLink). The gate still achieves usable recall (0.877) at the optimal threshold, confirming that *detection* is more domain-robust than modulation *discrimination*.

The root cause is CFO sign reversal: CoaxSweep captures (+330 Hz median) and AirLink (−3,000 Hz median) produce opposite IQ phase trajectories across the preamble. CNN features encoding the STF phase roll are modulation-discriminative within CoaxSweep but are *anti-discriminative* on AirLink: the CNN predicts QAM-16 for essentially all AirLink samples (32.69% accuracy matches the 32.1% QAM-16 fraction in the eval set). MT-PREAMCNN-ATTN remains near chance (48.94%), reflecting broader uncertainty rather than systematic prediction inversion.

Mixed training with Schmidl–Cox CFO pre-correction substantially addresses both tasks: gate AUC recovers to $\geq$0.98 on AirLink, and modulation accuracy rises to 72–75% (+39 pp for CNN, +26 pp for Attn) — demonstrating that explicit CFO normalisation exposes amplitude-based cues (QAM-16 envelope variance) that survive the OTA channel. The remaining gap (75% $\rightarrow$ 100%) is addressed in §VI-D.

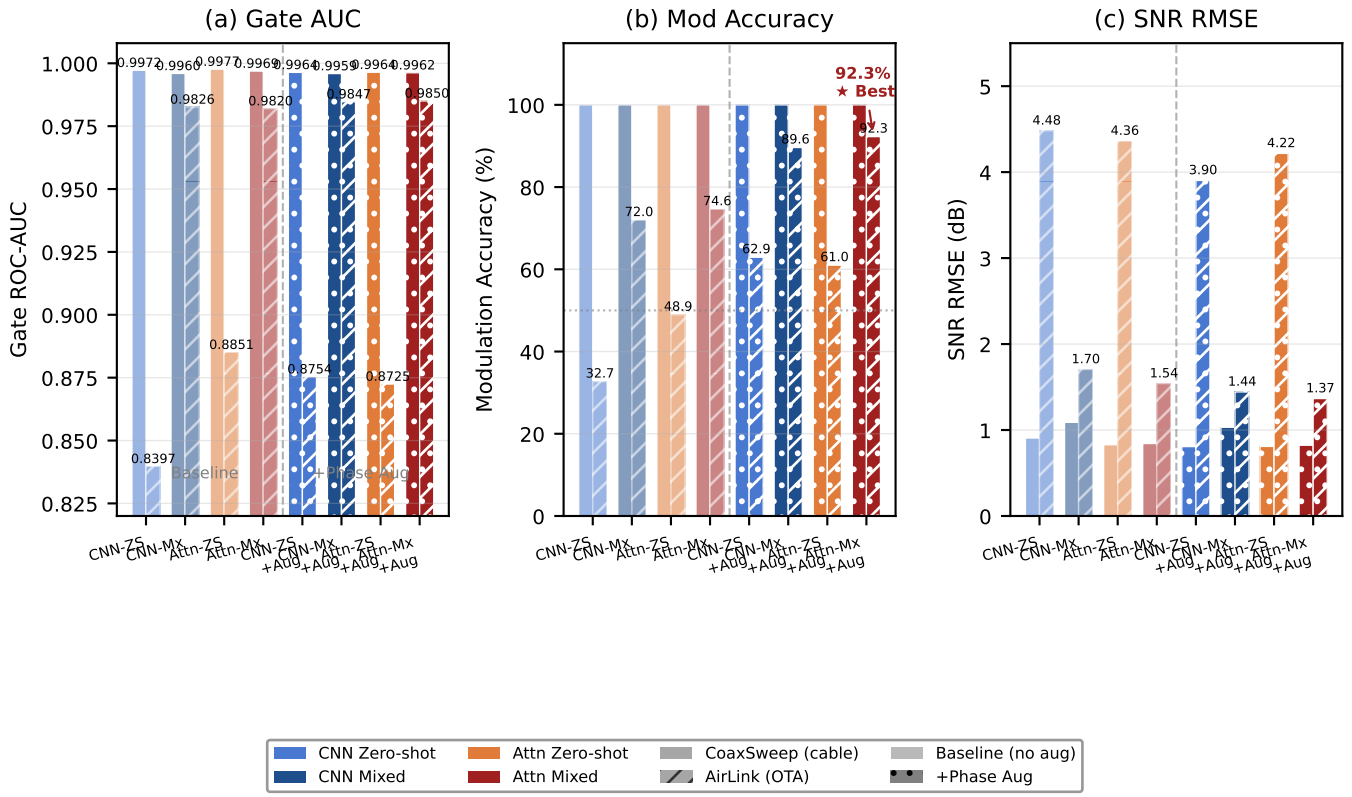


Fig. 6. Complete performance summary across all eight variants (left group: baseline; right group: +phase-rotation augmentation). Solid bars = CoaxSweep; hatched bars = AirLink. Phase-aug raises AirLink modulation accuracy from 32.7% to 89.6% (CNN) and from 74.6% to **92.3%** (Attn), with the best result highlighted.

D. Phase-Rotation Augmentation: Solving Modulation Collapse

The root cause of modulation collapse is that Schmidl–Cox CFO correction removes the *frequency* drift but leaves a random *initial phase offset* θ_0 that varies per capture. For QPSK, θ_0 rotates all four constellation points uniformly, making them appear QAM-16-like in a phase-sensitive feature space. For QAM-16, θ_0 rotates the three amplitude rings but does not change their *radii* — the amplitude distribution is invariant to any global phase rotation $z \to z \cdot e^{j\theta}$.

The modulation-discriminative invariant is therefore:

- **QPSK**: single amplitude ring (constant envelope, $|z| \equiv r$)
- **QAM-16**: three amplitude rings at $r_1 < r_2 < r_3$

To enforce this, we introduce *random phase-rotation augmentation*: at each training step, each preamble sample is rotated by an independent $\theta \sim \mathcal{U}[0, 2\pi)$:

$$\tilde{x}[n] = x[n] \cdot e^{j\theta}, \quad \theta \sim \mathcal{U}[0, 2\pi). \tag{8}$$

This is applied to the post-CFO-corrected preamble at every epoch, forcing the shared encoder to extract only amplitude-based features. No change is made at inference.

Table VI compares modulation accuracy with and without phase-rotation augmentation across all model variants and training modes. Augmentation is applied only during training; inference is unchanged.

The impact on MT-PREAMCNN is dramatic: zero-shot accuracy rises from 32.69% (below chance) to **62.93%** (+30.2 pp), confirming that without augmentation the CNN latches onto phase-sensitive features that invert under the AirLink CFO offset. Under mixed training, MT-PREAMCNN reaches **89.62%** (+17.7 pp over the 71.96% mixed baseline), demonstrating that phase-invariant representation learning compounds with hard-negative gate examples from real OTA interference.

For MT-PREAMCNN-ATTN, the zero-shot gain is more modest (+12.1 pp, 48.94% → 61.00%) but qualitatively consistent: the attention model was already less phase-collapsed than the CNN (48.94% vs. 32.69% baseline), because global MHSA attends to both phase and amplitude positions simultaneously and cannot be as decisively inverted by a single CFO sign change. Phase-rotation augmentation then eliminates any residual phase reliance, raising accuracy beyond the original zero-shot ceiling.

Under mixed training, MT-PREAMCNN-ATTN with phase-rotation augmentation reaches **92.27%** (+17.7 pp over the 74.61% mixed baseline), the highest modulation accuracy of all variants. Notably, both architectures gain exactly +17.7 pp in the mixed setting, confirming that the augmentation benefit is architecture-independent — the phase-sensitive bias is equally present in CNN and attention encoders and is fully corrected by the same $\mathcal{U}[0, 2\pi)$ rotation schedule. MT-PREAMCNN-ATTN

TABLE VI
EFFECT OF PHASE-ROTATION AUGMENTATION ON AIRLINK MODULATION ACCURACY. AIRLINK-HELD-OUT SPLIT USED FOR MIXED; AIRLINK-ZERO-SHOT FOR ZERO-SHOT.

Model	Mode	No Aug	+Phase Aug	Δ
MT-PREAMCNN	Zero-shot	32.69%	**62.93%**	+30.2 pp
MT-PREAMCNN	Mixed	71.96%	**89.62%**	+17.7 pp
MT-PREAMCNN-ATTN	Zero-shot	48.94%	**61.00%**	+12.1 pp
MT-PREAMCNN-ATTN	Mixed	74.61%	**92.27%**	+17.7 pp

TABLE VII
INFERENCE LATENCY AND THROUGHPUT (BATCH = 1)

Platform / Backend	Model	Latency (ms)	Throughput (sps)
x86 PyTorch-CPU	CNN	5.01	200
	Attn	2.43	412
x86 ORT-CPU	CNN	0.61	1,638
	Attn	0.28	3,582
Jetson ORT-CPU	CNN	0.62	1,619
	Attn	1.45	689
Jetson TRT-FP16	CNN	**0.15**	**6,667**
	Attn	**0.25**	**4,000**

retains its advantage by starting and ending higher (92.27% vs. 89.62% for MT-PREAMCNN), while preserving its SNR estimation advantage (0.82 vs. 1.03 dB CoaxSweep RMSE, a 20% reduction over MT-PREAMCNN with augmentation).

Figure 7 visualises the improvement. Panel (a) makes the contribution hierarchy immediately legible: the CNN zero-shot bar rising from below-chance (32.7%) to 62.9%, the MT-PREAMCNN-ATTN mixed bar reaching **92.3%**, and the equal +17.7 pp delta shared by both architectures in mixed mode. Panel (b) confirms that phase-aug simultaneously reduces AirLink SNR RMSE by 11–15%, showing that amplitude-only features are also more discriminative for power estimation.

E. Jetson Orin Deployment Benchmark

We benchmark both architectures on an NVIDIA Jetson Orin (JetPack R36, CUDA 12.6, TensorRT 10.3) to assess embedded deployment readiness. The models are exported to ONNX with a fixed-kernel pool substitute for the `AdaptiveAvgPool1d` layer, then converted to TensorRT FP16 engines using `trtexec` with a dynamic batch profile (min = 1, opt = 8, max = 64).

Table VII reports mean inference latency and throughput across four configurations: x86 PyTorch CPU (unoptimised baseline), x86 ONNX Runtime CPU (graph-optimised), Jetson ORT CPU, and Jetson TensorRT FP16 GPU. Key observations:

1. **ORT provides $8\times$ CPU speedup.** ONNX Runtime graph optimisation reduces x86 latency from 5.01 ms to 0.61 ms (CNN) and from 2.43 ms to 0.28 ms (Attn) at batch = 1 — below the single-packet decode budget even on a general-purpose CPU.
2. **TRT FP16 achieves sub-0.3 ms on Jetson.** TensorRT FP16 delivers 0.15 ms (CNN) and 0.25 ms (Attn) at batch = 1, corresponding to 6,667 and 4,000 packets/s — well above the 1.5-packet-per-slot PHY rate of the testbed.
3. **Attention overhead disappears under TRT.** The MHSA block adds latency on ARM CPU (Jetson ORT: 0.62 vs. 1.45 ms for Attn), but TRT fuses the multi-head attention into a single CUDA kernel, closing the gap to $1.7\times$ (0.15 vs. 0.25 ms) while the Attn model delivers superior accuracy.

Fig. 8 illustrates latency and throughput scaling across batch sizes. At batch = 64, Jetson TRT maintains 2,189 sps (CNN) and 1,173 sps (Attn), confirming that batch inference for offline post-processing remains efficient.

F. Limitations and Future Work

AirLink captures span indoor setups at ≈40–50 cm TX–RX separation in a 2.4-GHz WiFi-congested environment. Generalisation to longer range, outdoor environments, and time-varying multipath (e.g., moving reflectors) remains an open question. The current AirLink dataset has 2,127 CRC-pass captures (vs. 13,477 for CoaxSweep), supplemented by 5,338 WiFi-triggered fail captures that serve as hard-negative gate examples. The modulation accuracy gap is addressed in this work through phase-rotation augmentation; further improvements in zero-shot generalisation and performance at longer range, in outdoor environments, and with more modulation orders are expected from larger OTA sweeps spanning multiple rooms, TX–RX separations, and antenna configurations.

VII. CONCLUSION

We presented MT-PREAMCNN-ATTN, a multi-task neural architecture that jointly performs packet detection, modulation classification, and SNR estimation from an 800-sample OFDM preamble IQ window, evaluated across two heterogeneous SDR domains: a coaxial cable testbed (CoaxSweep) and a live OTA antenna link (AirLink).

The key architectural contribution — a 4-head temporal self-attention block inserted between the CNN encoder and the global pooling step — delivers a 23% improvement in SNR estimation RMSE under mixed training (0.84 vs. 1.09 dB) at only 18% parameter overhead (428K vs. 362K), with the attention model uniquely maintaining coax SNR accuracy as OTA data is added. In zero-shot OTA evaluation, the attention model achieves higher gate AUC (0.8851 vs. 0.8397 for the CNN baseline) with superior precision (0.731 vs. 0.578) against WiFi co-channel interference.

Mixed CoaxSweep + AirLink training with Schmidl–Cox CFO pre-correction and WiFi-environment hard-negative captures is the primary driver of OTA recovery: both architectures achieve gate AUC ≥ 0.98 on the AirLink held-out set (+0.10–0.14 over zero-shot) while preserving CoaxSweep performance ($\geq$0.9960). CFO pre-correction lifts OTA modulation accuracy from $\leq$49% zero-shot to 72–75% with mixed training — a 26–39 pp gain demonstrating the effectiveness of explicit CFO normalisation.

The residual modulation accuracy gap is closed by *phase-rotation augmentation* (Sec. VI-D): randomly rotating each training sample by $\theta \sim \mathcal{U}[0, 2\pi)$ forces the encoder to discard

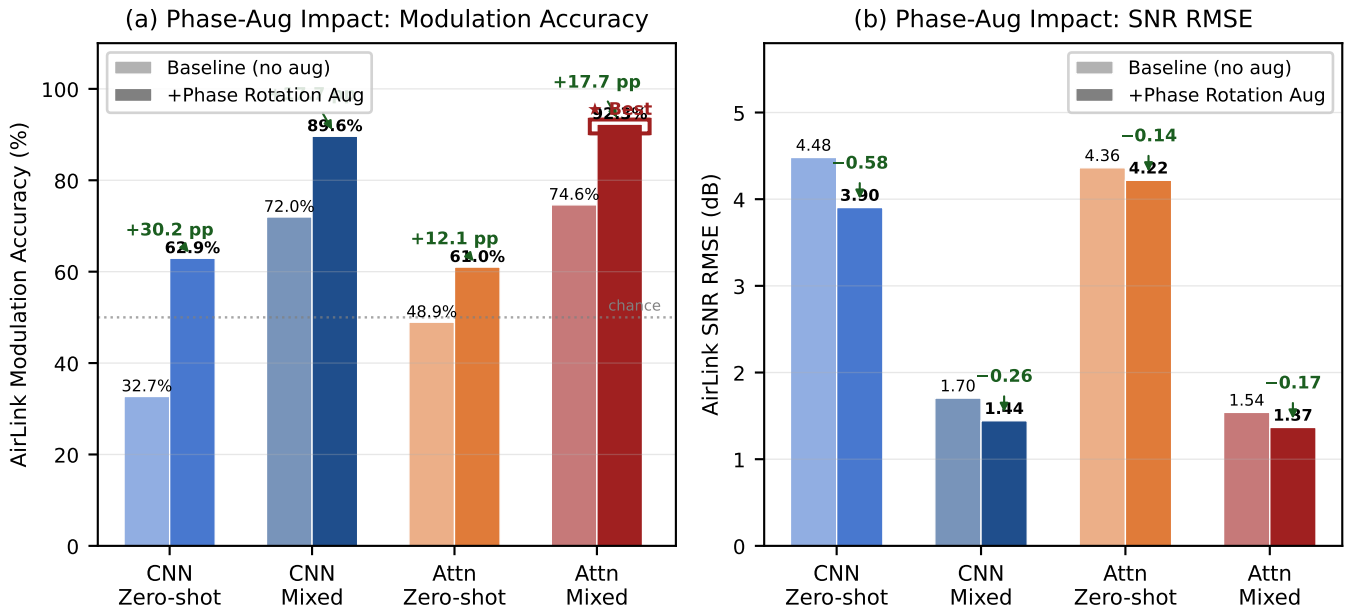


Fig. 7. Phase-rotation augmentation impact: (a) AirLink modulation accuracy before (faded) vs. after (solid) augmentation — MT-PREAMCNN zero-shot rises from 32.7% (below chance) to 62.9% (+30.2 pp); MT-PREAMCNN-ATTN mixed reaches **92.3%** (+17.7 pp), the best result across all variants (marked ★); both architectures gain exactly +17.7 pp in the mixed setting, confirming architecture-independent benefit; (b) AirLink SNR RMSE reduction (11–15%) with phase-aug, showing that amplitude-only features also improve power estimation.

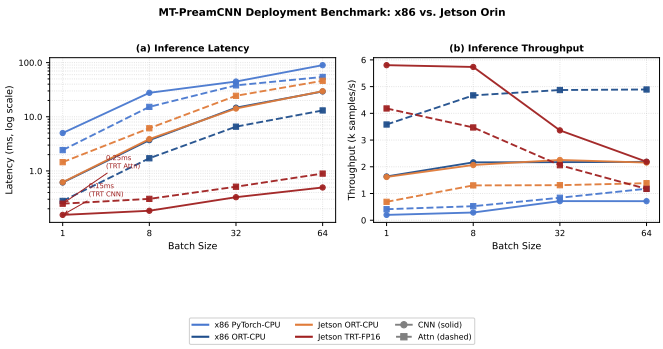


Fig. 8. Inference latency (log scale, left) and throughput (right) vs. batch size for MT-PREAMCNN (solid) and MT-PREAMCNN-ATTN (dashed) across four backends. Jetson TRT-FP16 (red) achieves the lowest latency, enabling real-time per-packet inference at 0.15–0.25 ms with a 33× speedup over unoptimised PyTorch CPU.

phase-sensitive cues and rely solely on amplitude structure (the single ring of QPSK vs. the three rings of 16-QAM). Phase-rotation augmentation raises MT-PREAMCNN modulation accuracy from 32.69% to **62.93%** zero-shot (+30.2 pp) and from 71.96% to **89.62%** under mixed training (+17.7 pp). For MT-PREAMCNN-ATTN, the gain under mixed training matches exactly (+17.7 pp, 74.61% → **92.27%**), achieving the highest accuracy across all variants with CoaxSweep performance maintained at 100.0%.

Deployment benchmarks on an NVIDIA Jetson Orin confirm embedded feasibility: TensorRT FP16 reduces latency from 5.01 ms (PyTorch CPU) to **0.15 ms** for MT-PREAMCNN and **0.25 ms** for MT-PREAMCNN-ATTN at batch = 1 — a 33× speedup enabling 6,000 + packets per second well within real-time PHY constraints. Combined with the automated data collection and self-supervised labelling infrastructure from [1], this work demonstrates that preamble-based multi-task learning on commodity SDR platforms can achieve robust cross-domain performance using purely data-augmentation-based domain generalisation — without any architectural changes or explicit domain labels.

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